

Post-retained Restorations: A Cost-minimization Analysis Nested in a Randomized Clinical Trial

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Clinical Relevance

Although the clinical performance of restorations was similar, the combination of glass fiber post and direct composite presented superior outcomes for cost versus other restorative possibilities for endodontically treated teeth.

INTRODUCTION

To achieve efficiency in the provision of health services, economic evaluations are necessary for decision making and improved allocation of resources. Economic health assessments aid in the organization of information in an understandable and simplified way; they can be defined as the comparison intervention options with the consideration of costs and consequences. Different types of economic analysis are used to represent health problems analytically without changing their essential attributes.^{1,2}

Cost-minimization analysis enables the evaluation of health intervention results and is imperative to inform and aid health policy decision making.²⁻⁴ It involves the performance of a prospective economic evaluation nested in a clinical trial that has revealed no significant difference in the primary clinical outcome,¹ thereby enabling identification of the intervention with the lowest cost.

When considering endodontically treated teeth and treatment choices, there is still a low body of evidence suggesting that cast metal posts have higher survival rates for teeth without ferrules⁵; however, all posts perform similarly when at least one remaining wall or a ferrule effect is present.^{6,7} Thus, in the presence of one or more remaining coronal walls, other parameters, especially the intervention cost, must be considered. The use of glass fiber posts (GFPs) is common and less expensive than the use of cast metal posts, which can influence clinical decision making. Glass fiber posts use less-expensive restorative materials and operative techniques, as adhesive approaches such as those for direct composite resin restoration reduce the amount of tooth structure removed and can extend the lifetimes of restored teeth, thereby avoiding more complex and expensive treatments.⁸ Also, as minor repairs of restorations are often required, economic evaluation in these cases can be critical to support clinical practice.

Given the amount of evidence regarding this type of analysis in dentistry and considering primary data from a randomized clinical trial (RCT), this study was conducted to economically assess four post-retained restorative strategies for endodontically treated teeth using cost-minimization analysis. The hypothesis tested was that glass fiber-retained direct composite resin restorations would be less expensive than restorations retained with cast post and core (CPC) or glass fiber posts with metal-ceramic crowns.

METHODS AND MATERIALS

This health economic evaluation was based on primary data from a parallel-arm RCT comparing success and

survival for four restorative strategies for endodontically treated teeth with and without ferrule use (ClinicalTrials.gov NCT01461239). The RCT was approved by the local research ethics committee (protocol 122/2009). The reporting of this study followed the Consolidated Health Economic Evaluation Reporting Standards (CHEERS).⁹

Target Population and Setting

The RCT was conducted at a dental school; 25 undergraduate students performed all clinical procedures under the researchers' supervision. One-hundred and forty-one participants were included, and 225 teeth were restored.

The inclusion criteria were good oral health (no active carious lesion or periodontal disease), the presence of one or more endodontically treated anterior or posterior teeth with or without ferrules (2 mm or 0-0.5 mm) requiring intraradicular retention, and the presence of bilateral posterior occlusal contact. Exclusion criteria were the presence of one or more endodontically treated teeth with periodontal or occlusal problem and the presence of a large prosthesis (Kennedy class I or II) in the opposing arch of the tooth or teeth to be restored.^{6,10}

Comparators and Sample Size

Randomization took the tooth type (anterior, molar, or premolar) into account to minimize bias due to anatomical variation. Briefly, when no ferrule was present, teeth were randomized according to the type of post (GFP, $n=71$; CPC, $n=79$) and all teeth received single metal-ceramic crowns. When a 2-mm ferrule was present, a GFP was luted, followed by a second randomization (metal-ceramic crown, $n=36$; direct composite resin restoration, $n=39$). Crowns were made with CoCr alloy^{6,10} and luted with self-adhesive resin cement (RelyX U100/U200; 3M ESPE, St Paul, MN, USA). The posts were luted with regular or self-adhesive dual resin cement (RelyX ARC or RelyX U100/U200; 3M ESPE). The randomization sequence was generated using computerized random number tables. A third party not involved in the study performed allocation concealment with the use of sealed opaque envelopes.

Under the assumption of no difference between treatments, 64 participants were required to achieve 90% confidence that the limits of a two-sided 90% confidence interval (CI) would exclude a difference $>18\%$ between the treatment groups.^{6,10} Considering the sample size for the GFP + CR⁺ and GFP + MC⁺ groups, an alpha level of 0.05, and success rates of 97.14% and 74.63%, respectively, this study had 90% power to detect incidence rate ratios ≥ 1.3 .

Follow-up

Overall, 225 teeth in 141 patients were evaluated annually between 2009 and 2018, with an average follow-up duration of 4.5 years (95% CI, 4.2-4.8).

Cost Estimation

The perspective of the Brazilian Public Health System (PHS) was adopted, considering total treatment coverage, as cost estimates accrue from primary and secondary care provided free of charge to patients. Costs were converted to US dollars (US\$1:R\$3.866). Prices were obtained in November 2018. An annual discount rate of 6.5% was applied to account for time preference.¹¹

The PHS procedure codes and national price lists were used to estimate direct medical costs, including initial treatment and maintenance (incremental) costs. Initial costs comprised those of all materials, consumables, dental technicians' work, and appointments required for the completion of a treatment. Incremental costs accrued from new interventions performed on teeth presenting repairable or irreparable failures. Material and consumable costs were accrued from the quantities (weights or amounts) of materials and their purchase costs, based on Brazilian PHS catalogs. Dental technician costs were in agreement with national standards for private practice. Incremental costs were calculated by summing the procedures required to repair a failure and/or replace a missing tooth. Non-medical costs (eg, structural costs, productive input) and indirect (opportunity) costs were assumed to be similar among treatments and were not considered.

Health Outcomes and Effectiveness

The primary health outcome was tooth extraction due to caries, root fracture, or endodontic complication. Associated costs included the extraction of the tooth and the insertion of an implant followed by a metal-ceramic implant-supported crown. The choice of this treatment was based on the inclusion criteria, justifying the replacement of the extracted tooth with a single implant in all cases of failure requiring tooth extraction. Secondary outcomes included the costs of repair, re-cementation, retreatment or apical surgery, and post/crown replacement. When restorative treatment or removal/replacement of the crown/restoration was performed at another dental practice (by another dentist) for no reconstructable reason, the affected tooth was censored and data from the last clinical assessment were considered.

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